

SANKET 2.0

Research & Documentation Pack

AI-assisted accessibility infrastructure for government service counters

Version 1.0 · 4 September 2026 · Team BeyondWords

Research Position

SANKET is positioned as counter-level accessibility infrastructure, not a generic ISL translator. The system combines immediate clerk assistance, clerk learning, institutional readiness measurement and human interpreter escalation.

Research Questions

The project studies constrained recognition, safe rejection, measurable validation, communication routing and integration of recognition into a clerk-first accessibility workflow.

Literature Synthesis

ISLRTC documents a 10,000-term dictionary with regional and contextual variation. ISL research includes image/CNN recognition, live video recognition, CNN/LSTM temporal approaches, continuous ISL-English resources and word-level datasets. These findings motivate SANKET's constrained vocabulary and explicit limitations.

Technical Baseline

The prototype uses MediaPipe Hand Landmarker, 21 hand landmarks, a normalized 42-dimensional representation, lightweight nearest-neighbour classification, quality gates, distance/margin rejection, voting and temporal stability.

Evaluation

The intended evidence workflow uses 20–50 real camera samples per sign, a deterministic 70/15/15 train/validation/test split, validation-only threshold calibration, held-out confusion matrices and negative/open-set testing.

Human-in-the-Loop

The design principle is: Technology when it can. Humans when it must. High-confidence predictions may be committed; uncertain observations should move toward confirmation, retry or interpreter escalation.

Ethics

Camera and dataset data should be minimized and collected with appropriate consent. The system should not make high-stakes administrative decisions from uncertain recognition. Accessibility status should not rely on color alone.

Limitations

The current prototype does not establish field accuracy, universal ISL translation, government deployment or official ISLRTC endorsement. Dynamic signs require temporal modeling, and sign assets require appropriate validation/provenance.

Research Contribution

The strongest defensible contribution is the system-level counter workflow and evidence framework around recognition: uncertainty, communication direction, learning, institutional measurement and human escalation.

Evidence Matrix — Core Claims

Claim	Evidence	Status
ISL Dictionary has 10,000 terms	ISLRTC FAQ	Supported
ISL has contextual/regional variation	ISLRTC FAQ	Supported
Continuous ISL-English datasets exist	ISLTranslate	Supported
Word-level ISL datasets exist	INCLUDE	Supported
Static recognition solves all ISL	No adequate evidence	Rejected
SANKET field accuracy	Requires field trial	Not established
Government deployment	No deployment evidence in prototype	Not established
Emoji are ISL	No	Rejected

Reference set

ISLRTC FAQ — <https://islrhc.nic.in/faq/>

ISLRTC ISL Dictionary — <https://islrhc.nic.in/isl-dictionary/>

Joshi, Agrawal & Modi (2023), ISLTranslate — <https://aclanthology.org/2023.findings-acl.665/>

ISLTranslate repository — <https://github.com/Exploration-Lab/ISLTranslate>

Rao et al. (2023), Image-based ISL Recognition review — <https://arxiv.org/abs/2304.14710>

IEEE: Indian Sign Language Detection Using Deep Learning — <https://ieeexplore.ieee.org/document/10581347/>

IEEE: CNN and LSTM approaches — <https://ieeexplore.ieee.org/document/10420379/>

IEEE: Accurate Gesture Recognition in ISL — <https://ieeexplore.ieee.org/document/11340772/>